

A Beginner's Guide To Debugging Docker Containers

Docker containers are great when they work, but frustrating when they exit immediately. Learning how to look inside a broken container is an essential skill for every junior engineer.

What you'll learn:

1. Check Why It Stopped With Docker Ps
2. Read The Final Logs For Clues
3. Peek Inside Using Interactive Shell
4. Inspect Deep Configs And Variables

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1. Check Why It Stopped With Docker Ps

When a container exits immediately, Docker saves a history of it. Running the 'docker ps -a' command shows you all containers, even the stopped ones. Look at the STATUS column to see if it exited with an error code like (1) or (137). This tells you if your app crashed due to a code error or ran out of memory.

```
$ docker ps -a
```

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2. Read The Final Logs For Clues

Before a container dies, it usually prints an error message to the console. You can view these messages using the logs command followed by your container ID or name. If the container keeps restarting in a loop, add the tail flag to see just the last few lines of error output.

```
$ docker logs --tail 50 <container_id>
```

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3. Peek Inside Using Interactive Shell

If the logs are not enough, you can step directly inside the container filesystem. By running 'docker run -it' with 'sh' or 'bash', you open a command prompt inside your container. This lets you check if your environment variables and files are in the right folders.

```
$ docker run -it <image_name> sh
```

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4. Inspect Deep Configs And Variables

Sometimes the issue is not your code, but how Docker is configured to run it. The inspect command dumps a massive JSON file containing every detail about your container network, volumes, and environment variables. You can search this output to confirm your settings are correct.

```
$docker inspect <container_id>
```

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Key Takeaways

- + Always use 'docker logs' as your very first debugging step
- + Name your containers using '--name' so they are easy to find
- + Keep your Dockerfiles simple to avoid complex startup errors
- + Test your container build locally before pushing to the cloud
- + Clean up old stopped containers regularly using 'docker container prune'

Watch Out For

- ! Assuming the container is running without checking 'docker ps'
- ! Panicking and rebuilding the image when it is just a missing environment variable
- ! Forgetting that container filesystems are wiped when the container is removed

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